The Clebsch hexagon code: a rigidity theorem for deep holes

An MDS code over $\mathbb{F}_{11}$ whose deep holes are exactly a conic

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Abstract

Among all six-arcs of the projective plane $\mathrm{PG}(2, 11)$, exactly one — up to the action of $\mathrm{PGL}(3, 11)$ — has the property that the deep holes of its associated MDS code lie on a conic. That arc is the *Clebsch hexagon*, the mod-11 reduction of the six axes of Klein's icosahedron (the six poles of an icosahedral $A_5 \subset \mathrm{PGL}_2(11)$ acting on a conic), and the conic condition alone recovers its full projective stabilizer A_5. The arc lies on no conic, so its $[6, 3, 4]_{11}$ code is projectively non-GRS of covering radius three; for this code the complete set of deep holes is not merely contained in a conic but is the entire set of $\mathbb{F}_{11}$-rational points of that conic, an identification recorded with its certificate in a companion paper and taken here as the starting point. We prove this rigidity as a five-way equivalence, establish a companion gap theorem quantifying its failure to be merely stable, exhibit a canonical automorphism-invariant chirality $\mathbb{Z}/2$ on the twenty weight-three coset leaders, and show that $q = 11$ is the only prime at which the configuration can occur *with icosahedral symmetry*: an elementary secant-counting bound forces $q \leq 14$, and $\mathbb{F}_q$-rationality of A_5 then leaves $q = 11$ alone. The underlying six-arc census and its extension-count spectrum belong to the classification of Hirschfeld and Sadeh; we recompute that spectrum and supply its coding-theoretic reading, but claim no priority on the numbers themselves.

1 Introduction

Let C be a linear $[n, k, d]_q$ code with covering radius ρ. A *deep hole* of C is a vector at distance exactly ρ from every codeword; equivalently, its coset in $\mathbb{F}_q^n/C$ has maximum possible weight among coset leaders. For MDS codes with a projective parity-check description — an arc A in $\mathrm{PG}(k-1, q)$ — there is a classical dictionary translating leader weights and covering radius into incidence statements about A: a syndrome direction has distance one, two, or three according as it lies on the arc, on a secant of the arc, or on neither; and adjoining such a point as a new arc column preserves the MDS property exactly when it is missed by every secant. This arc–coset dictionary, in various forms, underlies the classical theory of complete arcs and saturating sets and has been made precise and general for MDS codes by Davydov, Marcugini, and Pambianco [5], whose Theorem 6.3 gives the exact leader-count formula we use in Section 3. For plane sixth-degree ($[n, 3]$) codes the same dictionary is implicit already in the classical arc/covering-radius literature; see Hirschfeld [9]. For the standard (affine) Reed–Solomon setting, determining deep holes is a long-studied algorithmic and structural question; Guruswami and Vardy [8] initiated its decision-complexity study, showing maximum-likelihood decoding of Reed–Solomon codes to be NP-hard by essentially proving that deciding whether a received word is a deep hole is co-NP-complete, and Cheng and Murray [3] gave a substantially simpler proof of that same co-NP-completeness result and extended the study to standard (full-evaluation-set) Reed–Solomon codes. Zhang, Wan, and Kaipa [15] determine the deep holes of *projective* Reed–Solomon codes completely, identifying them with tangent lines to, and a quadratic-extension family over, the underlying normal rational curve (their Theorems I.4–I.7). Both of these are for the generalized Reed–Solomon (GRS) case, where the arc lies on a normal rational curve; the present code does not.

The object of this paper is a single, small, and highly symmetric exception to that GRS world: a $[6,3,4]_{11}$ MDS code, projectively non-GRS, whose covering radius is three and whose deep holes — rather than merely lying off the arc's fifteen secants, as they must for *any* incomplete arc — coincide exactly with the $\mathbb{F}_{11}$-points of a single nonsingular conic. The arc realizing this code is the *Clebsch hexagon*: six points of $\mathrm{PG}(2,11)$, projectively equivalent to the classical hexagon of Storme and Van Maldeghem [14] and studied synthetically over a general field by Dye [6], whose stabilizer in $\mathrm{PGL}(3,11)$ is the icosahedral group A_5.

That such an arc exists, and that its deep holes are a conic, is by itself a finite check — an instance of the DMP dictionary applied to one known arc. What we add is that this instance is not an accident of the arc chosen: among *all* six-arcs of $\mathrm{PG}(2,11)$ — fifteen classes under $\mathrm{PGL}(3,11)$, which our search meets as 1548 frame-normalized representatives — the Clebsch hexagon is the unique class whose deep-hole locus lies on any conic at all, degenerate or not, and this single containment condition already forces the stabilizer to contain A_5. We state this as a rigidity theorem (Section 4), quantify how sharply it fails to be a merely *stable* phenomenon (the companion gap theorem), exhibit a canonical chirality invariant on the code's weight-three coset leaders (Section 5), and explain why $q=11$ is the only prime at which the configuration can occur with icosahedral symmetry (Section 6): an elementary secant-counting bound caps q at 14, and rationality of A_5 closes the remaining range. We stress that the counting bound alone does not isolate $q=11$ — every prime power in range satisfies it — and that the rationality of A_5 enters there as a hypothesis on the arc, not as a consequence of the coding condition; the recovery of A_5 from the coding condition in Section 4 is a statement about $q=11$, not an input to the argument of Section 6. Section 7 discusses, without claiming any causal role for it, the sense in which these objects are the mod-11 reduction of Klein's icosahedral invariant theory.

What is new and what is not. The six-arc itself, its projective uniqueness, and its incompleteness at $q=11$ are due to Storme and Van Maldeghem [14] and, for its synthetic conic geometry, to Dye [6]. The classification of arcs (and in particular of six-arcs) in $\mathrm{PG}(2,11)$, together with the associated extension-count data, is due to Hirschfeld and Sadeh [10, 12]; see also the tables in Hirschfeld's book [9]. We recompute the extension-count spectrum of six-arcs as a byproduct of the rigidity theorem's exhaustive search, and we cite that classification for it rather than claim any priority on the numbers. The identification of this code's deep-hole locus with the A_5-invariant conic (Proposition 3.1) is recorded, with a kernel-checked certificate, in the companion arcs paper [1, Prop. 8.7]; we reprove it here to keep the present paper self-contained and claim no priority for it either. What we claim as new is the *reading* of that geometry: that the conic-containment condition characterizes the Clebsch hexagon among all six-arcs and recovers A_5 from a purely coding-theoretic hypothesis, not by construction; the quantitative gap theorem accompanying that rigidity; the canonical chirality invariant on the deep-hole coset leaders; and the argument isolating $q=11$ among arcs with icosahedral symmetry.

2 The Clebsch hexagon

Fix the nonsingular conic

$$\mathcal{C}: \; XZ = Y^2$$

in $\mathrm{PG}(2,11)$, with its 12 points identified with $\mathrm{PG}(1,11) = \mathbb{F}_{11} \cup \{\infty\}$ in the usual way. The stabilizer of $\mathcal{C}$ in $\mathrm{PGL}(3,11)$ is $\mathrm{PGL}_2(11)$, acting on $\mathcal{C}$ as it acts on the projective line.

Definition 2.1 (Clebsch hexagon)**.** Let $A_5 \subset \mathrm{PGL}_2(11)$ be an icosahedral subgroup. The *Clebsch hexagon* is the six-point orbit of poles of the six 5-fold-axis chords of A_5 acting on $\mathcal{C}$; equivalently, the six points of $\mathrm{PG}(2,11)$ each of whose two tangency lines to $\mathcal{C}$ meet $\mathcal{C}$ in an antipodal pair fixed by a Sylow 5-subgroup of A_5.

This is projectively the arc K_2 of Storme and Van Maldeghem [14, Prop. 11] at $q = 11$, and the hexagon studied synthetically by Dye [6] under the existence condition 5 a square in the base field. By [14, Prop. 12] it is the unique $\mathrm{PGL}(3,11)$-orbit of six-arcs with stabilizer A_5; by Dye's classification its five self-polar triangles are self-polar with respect to the same conic $\mathcal{C}$, and it lies on no conic.[1] The arcs of $\mathrm{PG}(n,q)$ fixed by a *primitive* A_5 or A_6 were catalogued more generally by O'Keefe and Storme [11], of which the present arc is the planar A_5 case (its stabilizer is 2-transitive on the six points, hence primitive); we cite that catalogue for completeness, and take the planar uniqueness statement we use from [14, Prop. 12]. Storme and Van Maldeghem's Proposition 13 records, by computer search, that this six-arc is incomplete at $q = 11$ (unlike at $q = 9$, where the analogous A_5-hexagon is complete); in the language of this paper, its covering radius as an arc is (at least) three.

We work throughout with the explicit representative

$$A = \{(1,10,0),\ (1,9,1),\ (1,4,7),\ (1,8,5),\ (0,1,4),\ (1,1,7)\},$$

already used in the companion arcs manuscript [1] for the same reasons: it is disjoint from $\mathcal{C}$, no three of its points are collinear, and its stabilizer in $\mathrm{PGL}(3,11)$ has order 60.

3 The code and its deep holes

Form the 3×6 matrix H whose columns are (representatives of) the six points of A, and let

$$C = \{c \in \mathbb{F}_{11}^6 : Hc^{\mathsf{T}} = 0\}$$

be the resulting $[6,3,4]_{11}$ MDS code. Since the six columns of H do not lie on any conic (their evaluation matrix against the six quadratic monomials $X^2, XY, XZ, Y^2, YZ, Z^2$ is nonsingular), C is projectively non-GRS: it is not the code of a normal rational curve, in the sense of the classical normal-rational-curve/GRS dictionary [13].

By the arc–coset dictionary of Davydov, Marcugini, and Pambianco [5, Def. 6.1, Thm. 6.3], a nonzero projective syndrome direction $x = [s]$ has coset-leader weight one, two, or three according as $x \in A$, x lies on a secant of A, or x lies on neither; in the last case the coset has exactly $\binom{n}{3}$ weight-three leaders, here $\binom{6}{3} = 20$. Their Theorem 6.3 is stated for an arbitrary n-arc of $\mathrm{PG}(2,q)$ and its associated $[n, n-3, 4]_q$ MDS code, with no Reed–Solomon, normal-rational-curve, or conic hypothesis and no restriction on q or the characteristic; it therefore applies to the projectively non-GRS code C at hand. (The GRS and conic specializations appear separately in their Theorems 6.4, 6.5, 6.8, and 6.10, which we do not

[1] Dye's paper is general-field synthetic geometry, and the $q = 11$ identification of the zero-bisecant locus with $\mathcal{C}$ — Proposition 3.1 below — does not appear among its stated results, per his own zbMATH summary (Zbl 0698.20032) and his 1997 self-recap; nor is it used by Storme and Van Maldeghem, who prove incompleteness at $q = 11$ by computer search without invoking any such fact from Dye. This is consistent with its structure: "$\mathcal{C}$ meets no bisecant" is a statement about $\mathbb{F}_q$-rational points, with no general-field content — over an algebraically closed field every bisecant meets $\mathcal{C}$, while over $\mathbb{R}$, where Dye's invariant conic $x^2 + y^2 + z^2 = 0$ has no points at all, the condition is vacuous; and already at $q = 9$ the Brianchon points lie on $\mathcal{C}$ — so it has no natural home in Dye's setting.

use. The leader count is also the $d = 4$, $n = 6$ case of the general $\binom{n}{d-1}$ of [5, Thm. 7.7], so it is an instantiation of a uniform formula rather than a coincidence at these parameters.) Finally, by their Definition 6.1 together with the definition of arc completeness, adjoining x as a seventh arc column preserves the MDS property exactly when x is missed by every secant.

Proposition 3.1 (Deep holes are the conic)**.** *The covering radius of C is three, and its projective deep-hole locus — the set of directions $x \notin A$ on no secant of A — is exactly the twelve points of $\mathcal{C}$.*

Proof. A direct finite computation over the 133 points of $\mathrm{PG}(2, 11)$: for each of the $\binom{6}{2} = 15$ secants of A, mark the $q - 1 = 10$ points it carries other than its two arc points. The marked set is $\mathrm{PG}(2, 11) \setminus (A \cup \mathcal{C})$ — all 115 points off the arc except the twelve points of $\mathcal{C}$, each of which has secant index zero. By the dictionary above, this uncovered locus is exactly the projective deep-hole locus, and its nonemptiness gives covering radius three rather than two. The computation is reproduced independently by `check_rigidity_degenerate_conic.py` (accompanying this paper) as the $|\mathcal{U}(A)| = 12$ entry of the census of Section 4. □

This identification, together with its coset data and a kernel-checked Lean certificate, is also recorded in the companion arcs paper [1, Prop. 8.7], where it arises as one instance of that paper's secant-index spectrum; we restate it here, with a self-contained proof, because it is the starting point of the rigidity theorem that is the present paper's subject. The results of Sections 4–6 — the rigidity equivalence, the gap theorem, the chirality invariant, and the isolation of $q = 11$ — appear in neither that paper nor elsewhere.

Corollary 3.2 (Deep holes as a named variety)**.** *Every affine deep hole of C lies over a projective direction on $\mathcal{C}$; the affine deep-hole count is $12 \times 10 = 120$, and each deep-hole coset has exactly $\binom{6}{3} = 20$ minimum-weight leaders. In particular, the complete deep-hole set of C is exactly the set of $\mathbb{F}_{11}$-points of the A_5-invariant conic $\mathcal{C}$: not merely contained in a positive-dimensional variety, nor built from one, but equal to its full $\mathbb{F}_q$-point set.*

Proof. Immediate from Proposition 3.1, the leader-count clause of the DMP dictionary, and the standard bijection between a projective syndrome direction and its $q - 1$ nonzero affine scalar multiples. □

The coset-leader-weight distribution of C is $(1, 60, 1150, 120)$ (cosets of weight $0, 1, 2, 3$, summing to $11^3 = 1331$), and its codeword weight enumerator is $(1, 0, 0, 0, 150, 420, 760)$, forced by the MDS parameters; we record it as a sanity check, not a novelty.

3.1 Certified structural facts

The following facts are either forced by the MDS parameters or are direct consequences of the arc's known geometry; we record them because they anchor the code's symmetry and design-theoretic reading, not as new results.

- **Automorphism group.** The permutation automorphism group of C (equivalently, the stabilizer of A in $\mathrm{PGL}(3, 11)$) has order 60, is isomorphic to A_5, and acts 2-transitively on the six coordinates — an exotic-hexad action of A_5 realized as a subgroup of S_6 via the outer automorphism of S_6. This is the natural $\mathbb{F}_{11}$/off-conic analogue of the hexacode $[6, 3, 4]_4$, whose automorphism group is the same exotic $A_5 \rtimes \cdot$ structure inside $S_6 \times \mathrm{GL}(1, 4)$; we cite the hexacode as precedent and do not claim the symmetry itself is unprecedented.

- **Design view.** The MDS parameters make C an orthogonal array $\mathrm{OA}(11^3, 6, 11, 3)$ of strength three; this is an automatic consequence of the $[6, 3, 4]$ parameters and is recorded here only to connect the code to the design-theoretic framing this venue expects.

- **Five self-polar triangles.** Dye's synthetic theory of the Clebsch hexagon [6] exhibits five triangles on the six vertices, each self-polar with respect to the unique conic $\mathcal{C}$ stabilized by the hexagon's automorphism group — the same conic as the deep-hole locus of Proposition 3.1. We attribute this structure to Dye and pose, without answering, whether the five triangles index a natural decomposition of the code's coset or weight structure.

4 The rigidity theorem

The content of Proposition 3.1 is, on its own, a single finite computation for one known arc. The following theorem is the paper's main claim: among *all* six-arcs of $\mathrm{PG}(2, 11)$, the property "deep holes lie on a conic" picks out the Clebsch hexagon uniquely, and does so in a way that recovers its automorphism group from the coding condition rather than assuming it.

Remark 4.1 (Two conditions, one word apart)*.* The condition studied here is that $\mathcal{U}(A)$ — the deep-hole locus — lies on a conic. It is *not* the familiar condition that the arc A itself lies on a conic, which is the standing hypothesis throughout the arc literature and which the Clebsch hexagon famously fails. The two are logically unrelated: they constrain different point sets, and the arcs satisfying them are disjoint. Indeed every six-arc lying *on* a conic in $\mathrm{PG}(2, 11)$ has $|\mathcal{U}(A)| \in \{18, 19, 20, 22\}$, so by (iii) none of them comes anywhere near the condition of the theorem.

Census framing. The projective-equivalence classes of k-arcs in $\mathrm{PG}(2, 11)$ were classified by Sadeh [12, 10]; see also Hirschfeld [9, Ch. 14]. Over the six-arc classes of that census we compute, for each representative A, the deep-hole locus $\mathcal{U}(A)$ — the points off A and off all fifteen of its secants, equivalently the points extending A to a projective seven-arc — and read $\mathcal{U}(A)$ as the covering-radius-three coset data of the associated $[6, 3, 4]_{11}$ code. *We claim no priority on the six-arc census, nor on the raw extension-count data $|\mathcal{U}(A)|$ that an extension-based classification entails*: the size of $\mathcal{U}(A)$ is a standard projective invariant of any arc, and the census itself is Sadeh's. What we contribute is the geometric and coding-theoretic meaning of $\mathcal{U}(A)$: that its containment in a conic singles out the Clebsch hexagon and forces A_5, that this containment is rigid rather than merely typical, and that for the singled-out class the complete deep-hole set is the full set of $\mathbb{F}_{11}$-points of a conic.

Theorem 4.2 (Rigidity theorem)**.** *For a six-arc $A \subset \mathrm{PG}(2, 11)$, the following are equivalent:*

(i) the deep-hole locus $\mathcal{U}(A)$ lies on some conic (possibly degenerate);

(ii) $\mathcal{U}(A)$ is exactly the set of $\mathbb{F}_{11}$-points of a nonsingular conic;

(iii) $|\mathcal{U}(A)| \leq 15$ (in fact, $|\mathcal{U}(A)| = 12$);

(iv) A is $\mathrm{PGL}(3, 11)$-equivalent to the Clebsch hexagon;

(v) the stabilizer of A in $\mathrm{PGL}(3, 11)$ contains A_5.

Proof sketch. Every four points of a six-arc, no three collinear, form a projective frame; normalizing that frame to $e_1, e_2, e_3, (1,1,1)$ and sweeping the remaining two points over all legal completions meets every projective class of six-arc at least once, giving 1548 frame-normalized representatives. The enumeration is by representatives, not by classes: a class with stabilizer S appears $360/|S|$ times among them, so the class count is $\sum_{\text{reps}} |S|/360 = 15$. Exhaustiveness is what the argument needs, and representatives supply it; the multiplicities are an artifact of the normalization and carry no projective meaning. For each representative, $\mathcal{U}(A)$ and its conic membership are computed by exact linear algebra (secant coverage by direct enumeration; conic containment by the nullspace of the 6×6 quadratic-monomial evaluation matrix restricted to $\mathcal{U}(A)$, allowing degenerate quadrics). This yields the extension-count histogram

$$|\mathcal{U}(A)| \in \{12, 16, 18, 19, 20, 21, 22\}, \qquad \text{with multiplicities } \{6, 30, 150, 300, 630, 360, 72\}$$

over the 1548 normalized representatives (summing to 1548); the multiplicities are an artifact of the frame-normalized enumeration, and the projective invariant is the value-set itself together with its per-class assignment, which we grant outright to the Hirschfeld–Sadeh classification. The minimum, $|\mathcal{U}(A)| = 12$, is attained by exactly six of the 1548 representatives, all in a single $\mathrm{PGL}(3,11)$-orbit (multiplicity $6 = 360/60$, consistent with a stabilizer of order 60); this orbit is the Clebsch hexagon class, and on it $\mathcal{U}(A)$ is exactly the twelve points of the A_5-invariant conic $\mathcal{C}$. No other class has $\mathcal{U}(A)$ contained in *any* conic — every one of the remaining classes, including the concyclic six-arcs (those whose own six points lie on a conic, 252 of the 1548 representatives), has $|\mathcal{U}(A)| \geq 16$ with $\mathcal{U}(A)$ off every conic. This gives (iii)$\Leftrightarrow$(iv)$\Leftrightarrow$(v), and (ii)$\Rightarrow$(i) is immediate. For (i)$\Rightarrow$(ii) we exclude the degenerate case, since condition (i) as stated permits a degenerate conic (a line-pair or double line): the conic-containment test above is the rank over $\mathbb{F}_{11}$ of the $|\mathcal{U}(A)| \times 6$ quadratic-monomial evaluation matrix, which detects *any* nonzero quadratic form vanishing on $\mathcal{U}(A)$, degenerate or not. The exhaustive computation shows that exactly the six Clebsch-class representatives have this rank below 6; for each, the kernel is one-dimensional and its quadratic form is nonsingular (nonzero 3×3 determinant), with $\mathcal{U}(A)$ equal to all twelve $\mathbb{F}_{11}$-points of that conic; and *no* six-arc has $\mathcal{U}(A)$ supported on a degenerate conic. Hence (i)$\Rightarrow$(ii), completing the equivalence.[2] $\square$

Corollary 4.3. *A_5 is recovered from the coding condition "deep holes lie on a conic," not assumed as a hypothesis.*

Remark 4.4 (Which implications are new). The equivalence (iv)$\Leftrightarrow$(v) is not ours and is not new: that the Clebsch hexagon is the unique $\mathrm{PGL}(3,11)$-orbit of six-arcs whose stabilizer contains A_5 is Storme and Van Maldeghem [14, Prop. 12], quoted in Section 2 above, and the classification of six-arcs that underlies it is Sadeh's. The content of the theorem is the implication (i)$\Rightarrow$(iv), and with it the equivalence of the deep-hole conditions (i)–(iii) with the classical characterizations (iv)–(v). It is that bridge — from a covering-radius property of the code to a projective classification — that is the claim, and the corollary above is its consequence.

The rigidity theorem leaves open how close a *different* six-arc can come to the phenomenon. The next theorem shows the answer is: not close at all.

[2] The degenerate-conic exclusion and the $|\mathcal{U}(A)|$ histogram were verified by an independent second enumeration (`check_rigidity_degenerate_conic.py`); it reproduces the histogram $\{12{:}6, 16{:}30, 18{:}150, 19{:}300, 20{:}630, 21{:}360, 22{:}72\}$ and returns exactly the six concyclic arcs, all nonsingular, none on a degenerate conic.

Theorem 4.5 (Gap theorem)**.** *The Clebsch configuration is rigid rather than merely stable. Every non-Clebsch six-arc has* $|\mathcal{U}(A)| \geq 16$ *with* $\mathcal{U}(A)$ *contained in no conic; and every one of the* 252 *six-arcs obtained from the Clebsch hexagon by moving exactly one of its points has symmetric difference* $|\mathcal{U}(A) \triangle \mathcal{C}| \geq 18$ *from the conic (so at most seven of the twelve conic points survive as deep holes), with the exact symmetric-difference spectrum* $\{18, 19, 20, 22, 24\}$*. The distance from "deep holes on a conic" to the nearest other six-arc jumps from zero to at least eighteen, with nothing in between.*

Proof. We verify by exhaustive computation over the 252 single-point perturbations of the Clebsch hexagon (each of the six points moved to each of the other 42 points of $\mathrm{PG}(2, 11) \setminus A$ that keeps the result a six-arc) that $|\mathcal{U}(A) \triangle \mathcal{C}|$ takes exactly the values listed, with minimum 18 and no perturbation landing $\mathcal{U}(A)$ on any conic. The first clause restates the exhaustive classification of Theorem 4.2. □

Both computations are, in their present form, exhaustive finite checks; a kernel-checked Lean formalization of the enumeration and the perturbation census is planned but not yet complete, and is not required for the statements above, which are verified by direct computation over $\mathrm{PG}(2, 11)$.

5 The chirality invariant

The six columns of A give $\binom{6}{3} = 20$ three-element subsets, in ten complementary pairs $\{S, S^{\mathsf{c}}\}$. The stabilizer A_5 acts on these ten pairs as the Petersen graph, with adjacency "share exactly two of the three columns," and acts on the twenty triples themselves as two complementary orbits of size ten: for every S, the orbit of S under A_5 never contains S^{c}.

Proposition 5.1 (Canonical chirality)**.** *The twenty weight-three coset leaders of* C *split into two complementary* A_5*-orbits of size ten, and no automorphism of* C *exchanges the two orbits:* $\mathrm{Hom}(A_5, \mathbb{Z}/2) = 0$*, so every element of the automorphism group* A_5 *acts as an even permutation on the twenty triples, while the sixty permutations that do swap the two orbits are exactly the odd elements of the exotic* $S_5 \subset S_6$ *realizing the outer automorphism. Consequently the two orbits carry a canonical, automorphism-invariant* $\mathbb{Z}/2$*-valued label on the deep-hole coset leaders — a certified non-identifiable latent variable, since it is fixed under all sixty symmetries of the code yet not a constant.*

Proof. The orbit and adjacency claims are a direct computation of the A_5-action on the $\binom{6}{3} = 20$ subsets of the six columns, using the explicit generators of the stabilizer computed from A. That $\mathrm{Hom}(A_5, \mathbb{Z}/2) = 0$ is standard, since A_5 is simple and nonabelian; hence any homomorphism from A_5 to $\mathbb{Z}/2$ is trivial, and no element of the stabilizer can act as an odd permutation on a two-element orbit set unless it lies outside A_5 in the full S_5 realized by the exotic S_6-outer-automorphism action on the six columns. The verified computation identifies exactly the sixty orbit-swapping permutations with this complementary odd coset. □

The twenty weight-three coset leaders are the twenty coset leaders associated with deep-hole projective directions; the chirality label is thus intrinsic to the code's covering-radius data, not an auxiliary combinatorial overlay.

6 Why $q = 11$

The phenomenon of Sections 3–4 is specific to $q = 11$. Two independent constraints meet only there: an elementary counting bound that caps q, and the rationality of the icosahedral group that selects within the capped range. We are explicit about the division of labour, because the counting bound is often the more memorable half and is the weaker one: it excludes no prime power in range on its own.

Lemma 6.1 (Secant-covering bound)**.** *Let A be a six-arc of* $\mathrm{PG}(2,q)$ *disjoint from a conic* $\mathcal{C}$*, and suppose A is complete outside $\mathcal{C}$: every point of* $\mathrm{PG}(2,q)$ *lying off A and off $\mathcal{C}$ lies on some secant of A. Then*

$$15(q-1) \;\geq\; q^2-6,$$

equivalently $q^2-15q+9\leq 0$, so $q\leq 14$.

Proof. Since A has six points and $\mathcal{C}$ has $q+1$ points, and A is disjoint from $\mathcal{C}$,

$$|\mathrm{PG}(2,q)\setminus(A\cup\mathcal{C})| = (q^2+q+1)-6-(q+1) = q^2-6.$$

Each of the $\binom{6}{2}=15$ secants of A is a line of $\mathrm{PG}(2,q)$, hence has $q+1$ points, exactly two of which lie in A; so each secant covers at most $q-1$ points of $\mathrm{PG}(2,q)\setminus A$, and in particular at most $q-1$ points of $\mathrm{PG}(2,q)\setminus(A\cup\mathcal{C})$. Completeness outside $\mathcal{C}$ says the fifteen secants together cover all q^2-6 points of this set, so

$$q^2-6 \;\leq\; 15(q-1).$$

(No accounting for overlap between secants is needed here: the bound only uses that fifteen sets of size at most $q-1$ cover a set of size q^2-6, which requires the sum of their sizes to be at least q^2-6 regardless of how much they overlap; overlap can only make the true covered count smaller than the sum, never larger, so the inequality direction needed for an upper bound on q is exactly the one overlap cannot violate.) Rearranging, $q^2-15q+9\leq 0$, whose real roots are $\frac{15\pm\sqrt{189}}{2}\approx 0.62, 14.38$; since q is a positive integer, $q\leq 14$. □

Theorem 6.2 (Uniqueness of $q=11$)**.** *Suppose q is a prime power such that some six-arc of* $\mathrm{PG}(2,q)$ *is complete outside a conic $\mathcal{C}$ (every point off the arc and off $\mathcal{C}$ lies on a secant) and has deep-hole locus exactly the $\mathbb{F}_q$-points of $\mathcal{C}$. Then $q\leq 14$. If in addition the arc's stabilizer contains an icosahedral $A_5\subset\mathrm{PGL}_2(q)$, then $q=11$.*

Two remarks on what this does and does not say. First, the counting bound of Lemma 6.1 is not what isolates $q=11$: *every* prime power $q\leq 14$ satisfies $15(q-1)\geq q^2-6$ — at $q=13$, for instance, $180\geq 163$ — so the bound caps the range and nothing more. All of the discrimination inside that range is done by the rationality of A_5. Second, that rationality enters as a *hypothesis on the arc*, not as a consequence of the coding condition. This is worth stating plainly, because Theorem 4.2 recovers A_5 from the coding condition and it would be circular to read that as licence to assume A_5 here: the recovery is a theorem about $q=11$, established by an exhaustive search of $\mathrm{PG}(2,11)$, and carries no information about other q. Whether some six-arc with no icosahedral symmetry has deep-hole locus exactly a conic at another $q\leq 14$ is not settled by the argument below; we leave it open (Section 8).

Proof sketch. By Lemma 6.1, $q\leq 14$. Suppose the stabilizer contains an icosahedral $A_5\subset\mathrm{PGL}_2(q)$. By the classical Dickson criterion, $A_5\subset\mathrm{PSL}_2(q)$ forces $q\equiv\pm1 \pmod{10}$, or $q=5^h$, or $q=2^{2h}$ (where $A_5\cong\mathrm{PSL}_2(4)$); the prime powers at most 14 meeting one of these are $q=4,5,9,11$. At $q=4$ a six-arc of $\mathrm{PG}(2,4)$ is a hyperoval, hence complete, so its deep-hole locus is empty and the phenomenon is vacuous; the same holds at $q=5$, where six points exhaust $q+1$ and the arc is again complete. At $q=9=3^2$ the rationality filter passes ($9\equiv -1 \pmod{10}$), so $q=9$ must be excluded on its own terms. An icosahedral $A_5\subset\mathrm{PGL}_2(9)$ is unique up to conjugacy, and its orbits on the 91 points of $\mathrm{PG}(2,9)$ have sizes $6, 10, 15, 30, 30$ — none of size one or five. An A_5-invariant point set is a union of orbits, so the six-orbit is the *only* A_5-invariant six-set: every A_5-invariant six-arc of $\mathrm{PG}(2,9)$ is projectively the same one. It is a genuine six-arc, and it is complete — covering radius two, not three — so its deep-hole locus is empty rather than a conic and the phenomenon is vacuous at $q = 9$. Completeness is the classical computation of Storme and Van Maldeghem [14, Prop. 13]; we independently re-verified it (`check_q9_exclusion.py`), applying the same secant-coverage test used throughout this paper to the $q = 9$ witness arc frozen in the companion Lean development, and confirm $|\mathcal{U}(A)| = 0$. This leaves $q = 11$. $\square$

Three coincidences meet at 11 and nowhere else: it is the only prime with $p + 1 = 12$, so the twelve vertices of the icosahedron exhaust the projective line; it is prime to 60, so the icosahedral group survives reduction faithfully; and it is small enough that the fifteen secants of a six-arc have just enough room to cover every point off the conic. Only the last of these is the counting inequality, and it forces just $q \leq 14$; rationality of A_5 closes the door on the rest of that range. The failure past $q = 11$ is worth recording concretely, since the next admissible prime is the first place the construction could have continued. At $q = 19$ ($19 \equiv -1 \pmod{10}$, so the rationality filter passes and only Lemma 6.1 excludes it) the same recipe — the poles of the six five-fold axes of an icosahedral $A_5 < \mathrm{PSL}_2(19)$ — does still produce a genuine six-arc disjoint from the conic, so the object exists; what fails is the covering. We built that arc and enumerated its deep-hole locus directly (`check_q19_nonexample.py`), obtaining

$$|\mathcal{U}(A)| = 140 = 20 + 120$$

against a conic of only $q + 1 = 20$ points. The arithmetic differs from $q = 11$ in a way worth noting: there $5 \mid q - 1$, so an order-five element splits and fixes two rational points of $\mathcal{C}$, whereas at $q = 19$ we have $5 \mid q + 1$, so the order-five elements lie in a non-split torus and fix no rational conic point at all — their fixed pairs are conjugate over $\mathbb{F}_{361}$. The chord through such a pair is Galois-stable and hence still a rational line, so its pole is still a rational point and the arc is still defined; we verify this in the script rather than assume it.

The shape of the failure is instructive. The twenty conic points remain deep holes — $\mathcal{C} \subset \mathcal{U}(A)$ still — but they no longer exhaust $\mathcal{U}(A)$: 120 further points of $\mathrm{PG}(2,19)$ escape all fifteen secants, and $\mathcal{U}(A)$ lies on no conic (the quadratic-monomial evaluation matrix of $\mathcal{U}(A)$ has full rank six). The deficit itself is forced by Lemma 6.1: the fifteen secants have total capacity $15(q - 1) = 270$, while $q^2 - 6 = 355$ points lie off the arc and off the conic, so at least 85 points must escape, and 120 in fact do. (That the twelve — here twenty — conic points also remain uncovered is a separate fact, not something the lemma predicts.) At $q = 11$ the same capacity, 150, comfortably exceeds the 115 points needing cover, and the excess is spent on overlap. The phenomenon therefore degenerates immediately past $q = 11$ rather than persisting into a family — and it degenerates by the conic ceasing to be *all* of the deep holes, not by its ceasing to be deep holes at all.

7 The Klein reduction

This section is a discussion of where the Clebsch hexagon and its conic sit relative to Klein's classical invariant theory of the icosahedron. None of the following is used in, or needed for, the rigidity theorem, the gap theorem, the chirality proposition, or the uniqueness argument of Section 6; we present it because the coincidence is striking and because it explains, without asserting any causal mechanism, why an icosahedral object appears at all.

Klein's icosahedron carries three basic invariant binary forms in the vertex coordinates (z_1, z_2) of degrees 12, 20, and 30: the vertex form f, vanishing on the twelve vertices; the face form H, of degree 20; and the edge form T, of degree 30; these satisfy the classical syzygy $H^3 + T^2 = 1728\, f^5$. Reducing modulo a prime $\mathfrak{p} \mid 11$ (where 11 splits completely for the relevant field of definition), the icosahedral group $A_5 \subset \mathrm{PSL}(2, \mathbb{Q}(\zeta_5))$ reduces injectively to a conjugate of the arc stabilizer computed in Section 2. The vertex form f reduces to $z_1 z_2(z_1^{10} + 11 z_1^5 z_2^5 - z_2^{10})$, which modulo 11 collapses to $z_1 z_2(z_1^{10} - z_2^{10})$: the standard degree-$(q+1)$ Dickson invariant, vanishing on every point of $\mathrm{PG}(1, 11)$ and invariant under the full group $\mathrm{PGL}_2(11)$, not merely A_5 [4]. The six diagonals whose roots are enumerated by Klein's sextic resolvent reduce to the six chords whose poles are precisely the six points of the Clebsch hexagon; the syzygy $H^3 + T^2 = 1728 f^5$ reduces, since $1728 \equiv 1 \pmod{11}$, to $H^3 + T^2 = f^5$ on the corresponding mod-11 forms.

These objects are not analogues of Klein's icosahedral forms; they are their reductions. But the reduction is explicitly *non-causal* for the coding phenomenon of Sections 3–4: the vertex form f, once reduced mod 11, gains the full Dickson invariance of $\mathrm{PGL}_2(11)$ and no longer remembers the icosahedron on its own — the group must be carried alongside the form to recover A_5 specifically. Correspondingly, the rigidity theorem's covering-exactness property is not inherited from the Klein reduction in any family: repeating the same construction (poles of the axis chords of the analogous exceptional group) at the other members of the mod-p Platonic family — the tetrahedron at $p = 3$, the octahedron and cube at $p = 5, 7$, the dodecahedron at $p = 19$ — fails to reproduce the complete-outside-conic property at every other member; only the icosahedron at $p = 11$ is covering-exact. The reduction explains the presence of an icosahedral object at $q = 11$; it does not explain, by itself, why that object's deep holes fill a conic. We follow here the established genre of reducing a classical invariant-theoretic configuration modulo an exceptional prime dividing its automorphism group's order, as carried out for the Klein quartic and $\mathrm{PSL}(2, 7)$ by Elkies [7, §3.3].

8 Further remarks

An edge-level Schreier witness. Join two of the twelve points of $\mathcal{C}$ whenever the chord through them contains a point of the arc. Each such chord is a *uni*secant of A, meeting it in exactly one point — it could not be a secant, since by Proposition 3.1 no point of $\mathcal{C}$ lies on a secant of A at all; of the $\binom{12}{2} = 66$ chords of $\mathcal{C}$, thirty carry exactly one arc point and thirty-six carry none. The resulting graph on twelve vertices is the icosahedron: five-regular with thirty edges, and in fact isomorphic to the standard icosahedral presentation by an explicit vertex bijection. The thirty edges partition into six five-edge matchings, one for each point of the arc, consisting of the chords through that point; each such matching misses exactly one antipodal pair, the six missed antipodal chords are distinct, and adjoining them completes the six matchings to a one-factorization. The six antipodal chords are among the thirty-six *non*-edges: an antipodal chord carries no arc point at all. That is polarity rather than coincidence — by Definition 2.1 each arc point is the *pole* of an antipodal chord; a pole lies on its own polar only when it lies on $\mathcal{C}$, which no arc point does, and no two arc points are conjugate under that polarity, so no *other* arc point lies on it either. All of this is `decide`-grade and kernel-checked in the companion Lean development (the adjacency, the explicit isomorphism, the antipodal non-edges, and the one-factorization), and both the chord partition and the icosahedral identification are recorded in the companion arcs paper [1, Prop. 8.7(iv)]; we do not build any further claim on it here.

An open question: higher k. The present phenomenon is for plane arcs, i.e. codes of dimension $k = 3$. We do not know of any non-GRS MDS code of larger dimension whose complete deep-hole set is the $\mathbb{F}_q$-point set of a positive-dimensional named variety in the analogous sense; the twisted-cubic covering-code literature of Bartoli, Davydov, Marcugini, and Pambianco [2] gives the relevant $k = 4$ uncovered-locus machinery, but we pose the existence question for a codimension-four analogue of the present phenomenon as open, without a candidate construction. Whether a comparable phenomenon occurs for a different exceptional group or in higher dimension is likewise open: the natural dimension-up candidate, the Valentiner group $A_6 \subset \mathrm{PGL}_3(19)$, already fails at the first step, its smallest point orbit having $36 > q + 1$ points and so containing no arc.

Remark 8.1 (A labeled aside on statistical learning). The uniform twenty-way tie among weight-three coset leaders is a certified worst-case decoding instance achieving the Bayes-optimal error floor for any decoder on this code, and the chirality label of Proposition 5.1 is a certified example of a latent variable that is fixed under every symmetry of the instance yet not constant — a minimal symmetry-protected unlearnable bit. We flag this connection in one sentence and do not develop it further here.

The dual $[6, 3]$ code. The dual code $C^\perp$ of C is again a $[6, 3, 4]_{11}$ MDS code (since $n - k = 3 = k$ here), and its own parity-check matrix may be taken to be any generator matrix G of C; the six columns of G, read projectively, form the *dual arc* B. We computed G by exact nullspace reduction of H over $\mathbb{F}_{11}$ (`check_dual_code.py`) and found $B = \{(1, 5, 5), (1, 4, 9), (1, 9, 3), (1, 0, 0), (0, 1, 0), (0, 0, 1)\}$: a genuine six-arc (no three collinear), meeting the standard conic $\mathcal{C}$ in exactly two points, with deep-hole locus $\mathcal{U}(B)$ of size 12 lying exactly on a second, nonsingular conic $4X^2 + 7XY + 10XZ + 5Y^2 + 2YZ + Z^2 = 0$ — distinct from $\mathcal{C}$ as a point set. By the rigidity theorem (Theorem 4.2), $|\mathcal{U}(B)| = 12$ together with conic containment occurs for exactly one $\mathrm{PGL}(3, 11)$-orbit of six-arcs, so B is itself $\mathrm{PGL}(3, 11)$-equivalent to A: the dual of the Clebsch hexagon code is again a Clebsch hexagon code, for a different conic than the fixed standard one, not a structurally new arc. The rigidity phenomenon is thus self-dual, though not literally coordinate-for-coordinate self-dual.

The two icosahedral hexads and $S(5, 6, 12)$. The twelve points of $\mathcal{C}$ carry a Steiner system $S(5, 6, 12)$, built in the standard way as the orbit of the hexad $\{\infty\} \cup \mathrm{QR}(11)$ ($\mathrm{QR}(11) = \{1, 3, 4, 5, 9\}$, the nonzero squares mod 11) under $\mathrm{PSL}_2(11)$ acting by Möbius transformations on $\mathrm{PG}(1, 11) = \mathcal{C}$; we recomputed this orbit directly (`check_mathieu_hexads.py`) and confirmed it has the classical size 132 and the design property that every 5-subset of the twelve points lies in exactly one block. The six antipodal chords of Definition 2.1 partition the twelve conic points into two complementary size-six transversals, $\mathrm{hexad}A = \{0, 1, 2, 3, 5, 6\}$ and its antipodal complement $\mathrm{hexad}B = \{4, 7, 8, 9, 10, \infty\}$ (in the affine parametrization of Section 2). Neither transversal is one of the 132 Mathieu hexads: both meet every block in 1, 2, 3, 4, or 5 points, with the identical intersection-size histogram $\{1{:}6, 2{:}30, 3{:}60, 4{:}30, 5{:}6\}$, and in particular never in 0 or all 6. The two icosahedral hexads are transverse to $S(5, 6, 12)$ in the strongest sense available: they share substrate (the same twelve points) and a common subgroup ($\mathrm{PSL}_2(11)$ is a subgroup of M_{12}) with the Mathieu design, but are combinatorially unrelated to its blocks.

The ten-arc foil. Storme and Van Maldeghem's Proposition 11 gives, at $q = 11$, a second A_5-orbit six-and-ten-arc pair distinct from the hexagon. Rather than re-derive its explicit coordinates, we computed the stabilizer $\mathrm{Stab}(A) < \mathrm{PGL}(3, 11)$ of the Clebsch hexagon directly as sixty explicit 3×3 matrices (`check_ten_arc_foil.py`, verified $|\mathrm{Stab}(A)| = 60$), then let this concrete A_5 act on all 133 points of $\mathrm{PG}(2, 11)$. Its orbits have sizes

$\{6, 10, 12, 15, 30, 30, 30\}$ (the size-6 orbit is A itself, the size-12 orbit is $\mathcal{C}$); the unique size-10 orbit is

$$\{(1,0,3),(1,0,9),(1,3,3),(1,3,7),(1,5,4),(1,5,7),(1,6,4),(1,6,6),(1,7,9),(1,7,10)\},$$

a genuine ten-arc (no three collinear), disjoint from $\mathcal{C}$, with *empty* deep-hole locus. This is exactly the clean foil: the same icosahedral A_5 acting on the same plane produces, on a different orbit, a complete arc rather than one whose deep holes fill a conic — confirming that emptiness, not mere containment in a variety, is the generic covering-radius behavior, and that the Clebsch hexagon's exact conic-filling deep-hole locus is the exception rather than a typical outcome of icosahedral symmetry.

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